Machine Learning Project Report

Prepared by

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Python	Scikit-learn	XGBoost	SHAP	Power BI	Streamlit

February 2026 | IBM Telco Customer Churn Dataset | 7,032 Customers

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1. Executive Summary

This report presents a complete end-to-end Machine Learning project developed to predict customer churn for a telecommunications company. Using the IBM Telco Customer Churn dataset comprising 7,032 customer records, three machine learning models were trained and evaluated. The project delivers actionable business intelligence through predictive modeling, explainable AI using SHAP, an interactive Power BI dashboard, and a deployed Streamlit web application — enabling the business to proactively identify and retain at-risk customers before they leave.

7,032 Total Customers	1,869 Churned Customers	26.58% Churn Rate	$35,463 Revenue at Risk

■ **Best Model: Random Forest**	F1 Score: **0.6225**	ROC-AUC: **0.8338**	Recall: **0.7540**

2. Project Overview & Objectives

Customer churn — when a customer stops doing business with a company — is one of the most critical challenges in the telecommunications industry. Acquiring a new customer costs 5 to 7 times more than retaining an existing one. This project was built to solve a real business problem: **identify which customers are likely to churn so the business can intervene proactively.**

Business Problem

- The company is losing 26.58% of its customers — 1,869 out of 7,032
- Each churned customer represents lost monthly recurring revenue
- The business has no system to identify at-risk customers early
- Retention efforts are applied broadly rather than targeted at high-risk customers

Project Objectives

#	Objective	Outcome
1	Predict which customers will churn	Binary classification model (0=Stay, 1=Churn)
2	Identify key churn drivers	SHAP explainability analysis
3	Segment customers by risk level	High / Medium / Low risk segments
4	Quantify business impact	Revenue at risk: $35,463/month
5	Deliver actionable insights	Power BI dashboard + recommendations
6	Deploy real-time prediction tool	ChurnGuard AI — Streamlit web application

3. Dataset Description

The IBM Telco Customer Churn dataset is a widely used, real-world benchmark dataset in the data science community. It contains detailed information about customers of a fictional telecommunications company including their demographics, services subscribed, account information, and whether they churned.

Dataset Attribute	Details
Source	IBM Sample Data Sets — Telco Customer Churn
Total Records	7,043 customers (7,032 after cleaning)
Total Features	21 original features (30 after encoding)
Target Variable	Churn — Yes (1) / No (0)
Class Distribution	No Churn: 5,163 (73.42%) | Churn: 1,869 (26.58%)
Class Imbalance	Yes — addressed using SMOTE oversampling
Hidden Nulls	11 rows in TotalCharges (whitespace) — removed

Feature Categories

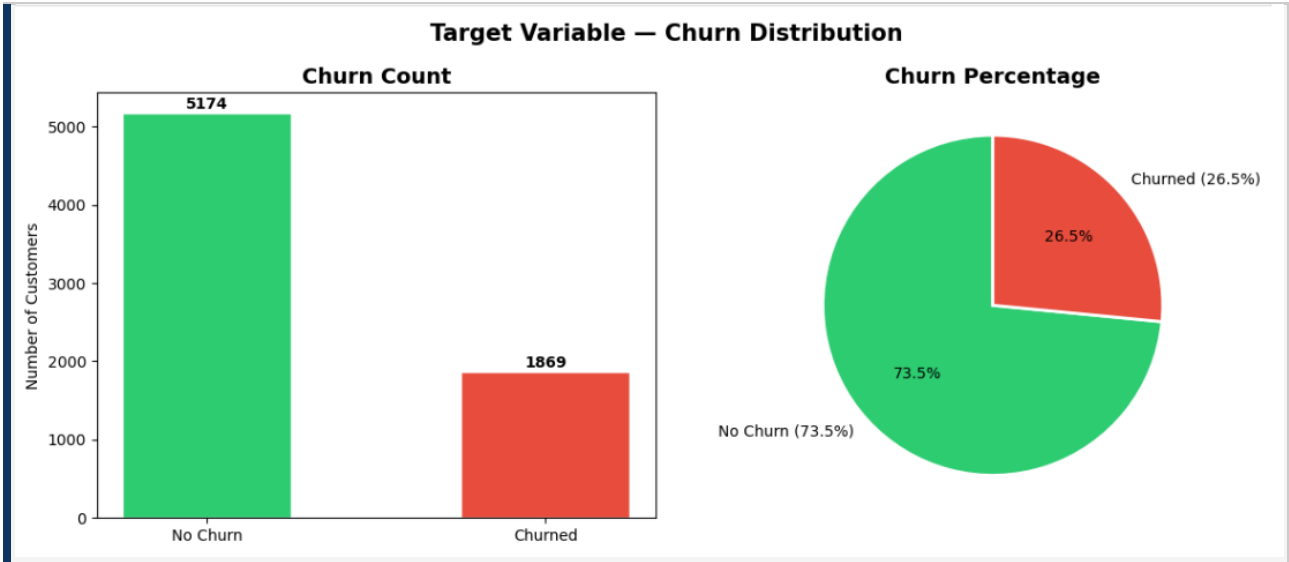
Category	Features
Demographics	gender, SeniorCitizen, Partner, Dependents
Account Info	tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Phone Services	PhoneService, MultipleLines
Internet Services	InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Target Variable	Churn (Yes/No)

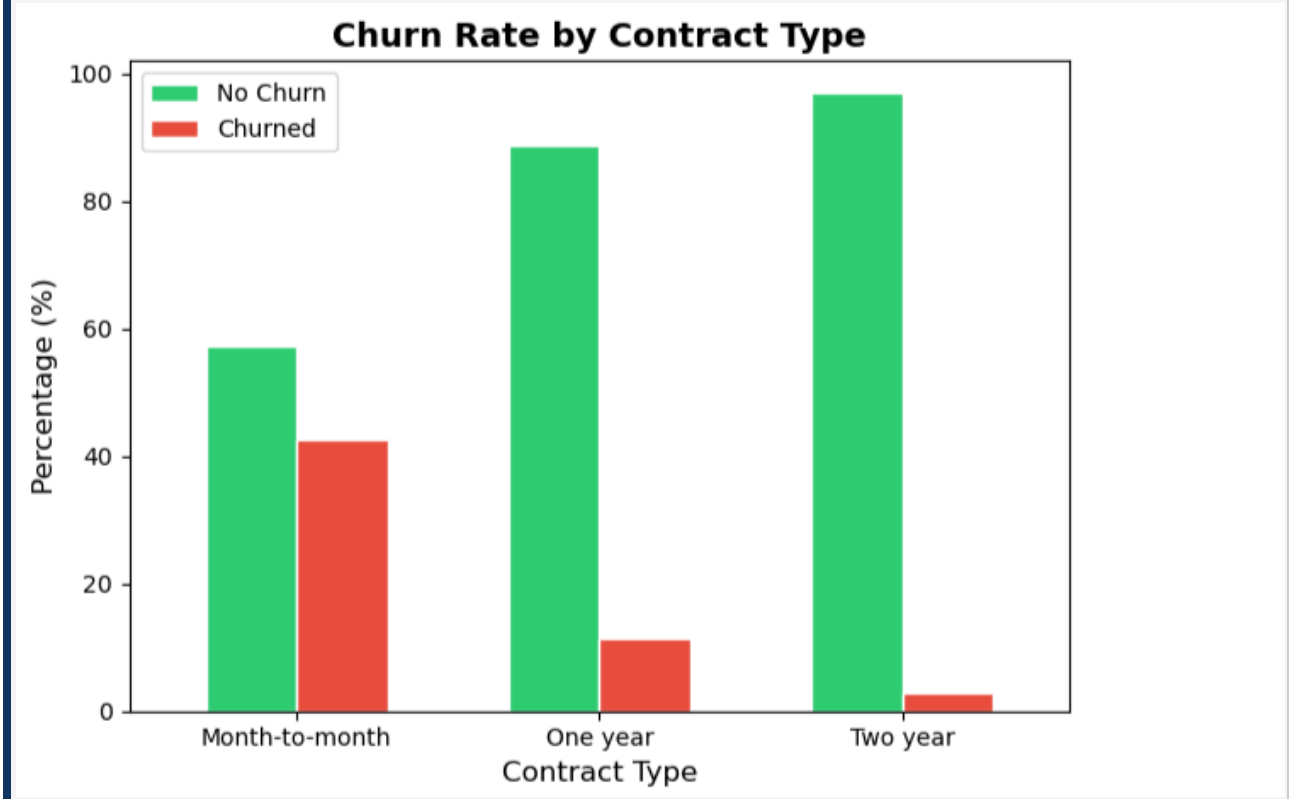
4. Exploratory Data Analysis (EDA)

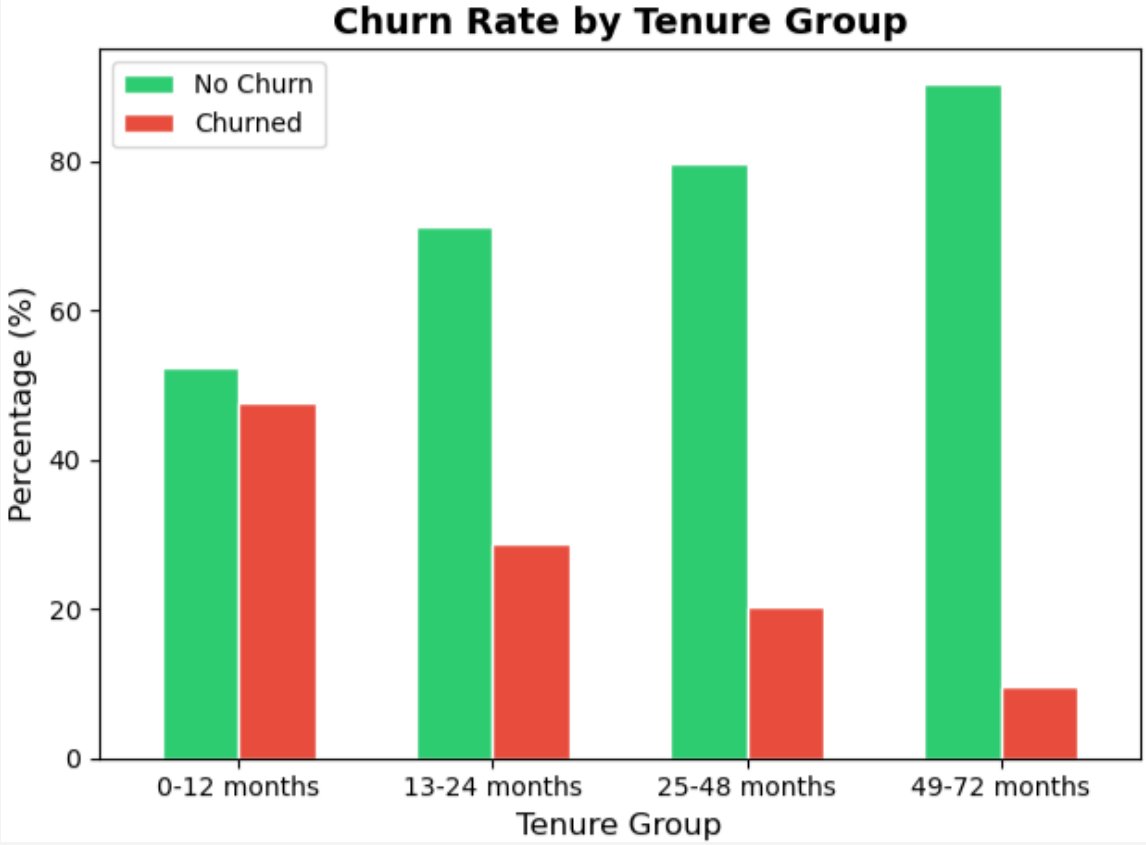
1	**Fix TotalCharges**	Converted TotalCharges from object (string) to numeric. Removed 11 rows with hidden whitespace nulls. Final dataset: 7,032 rows.
2	**Drop customerID**	Removed the customerID column as it is an identifier with no predictive value.

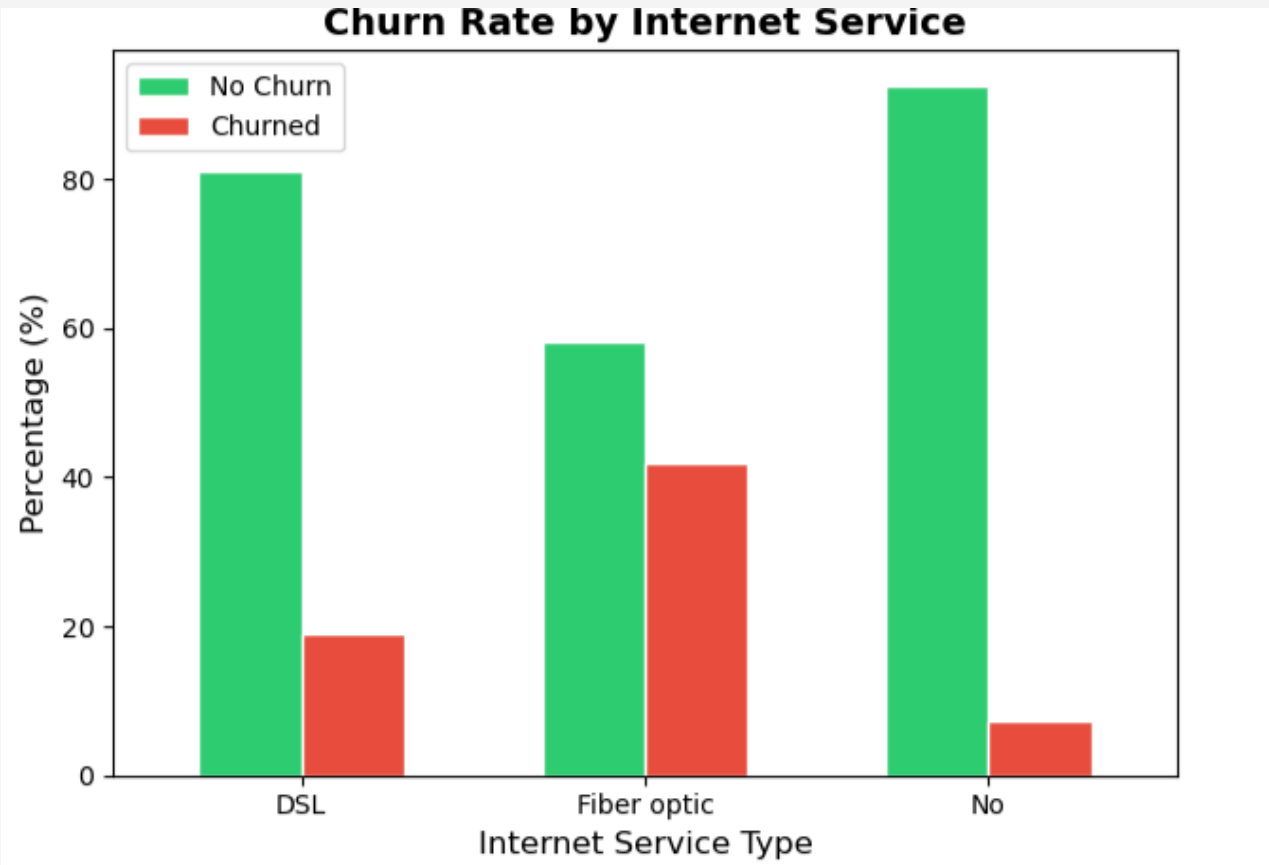
Exploratory Data Analysis was conducted to understand the structure, patterns, and relationships within the dataset before building any machine learning models. Six key visualizations were created to uncover the primary drivers of churn.

Chart	Finding
Churn Distribution	• 26.58% of customers churned — confirming significant class imbalance requiring SMOTE treatment.
Churn by Contract Type	• Month-to-month customers churn at ~42% vs only ~3% for two-year contracts — a 14x difference.
Churn by Tenure Group	• New customers (0-12 months) have 47% churn rate vs 8% for customers with 49-72 months tenure.
Churn by Internet Service	• Fiber optic users churn at 41% vs DSL at 19% — suggesting pricing or service quality issues.
Monthly Charges vs Churn	• Churned customers pay median $79/month vs $65/month for retained customers.
Correlation Heatmap	• Tenure (-0.35) is the strongest negative predictor; MonthlyCharges (+0.19) positive predictor.


Target Variable — Churn Distribution
Churn Count
Number of Customers
5174
1869
No Churn
Churned
Churn Percentage
Churned (26.5%)
26.5%
73.5%
No Churn (73.5%)


Churn Rate by Contract Type
No Churn
Churned
Percentage (%)
Month-to-month
One year
Two year
Contract Type


Churn Rate by Tenure Group
No Churn
Churned
Percentage (%)
0
20
40
60
80
0-12 months
13-24 months
25-48 months
49-72 months
Tenure Group


Churn Rate by Internet Service
No Churn
Churned
Percentage (%)
0
20
40
60
80
DSL
Fiber optic
No
Internet Service Type

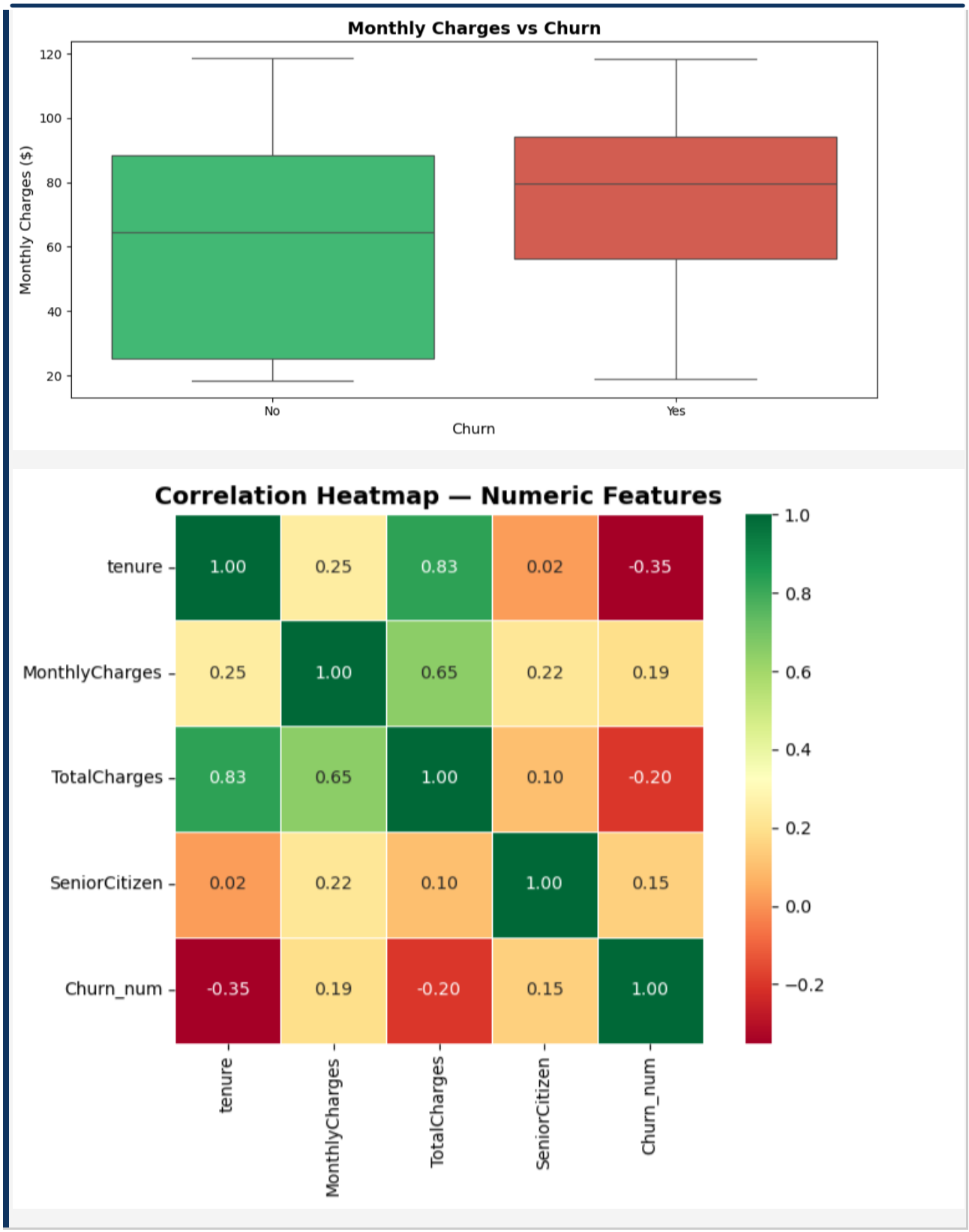


Figure 1: EDA Visualizations — Churn patterns across key features

5. Data Preprocessing

A structured 7-step preprocessing pipeline was implemented to prepare the data for machine learning. Each step addresses a specific data quality or modeling requirement.

3	**Encode Target Variable**	Mapped Churn column: Yes → 1, No → 0. Converted to integer type for compatibility with sklearn.
4	**One-Hot Encoding**	Applied pd.get_dummies() with drop_first=True to encode all 15 categorical features into 27 binary columns.
5	**Feature Scaling**	Applied StandardScaler to tenure, MonthlyCharges, and TotalCharges — ensuring equal weight in distance-based algorithms.
6	**Train-Test Split**	80/20 stratified split: X_train (5,625 rows), X_test (1,407 rows). Stratification preserves 26.58% churn ratio in both sets.
7	**SMOTE Oversampling**	Applied Synthetic Minority Oversampling Technique (SMOTE) to training data only. Balanced classes from 4,130 vs 1,495 to 4,130 vs 4,130.

SMOTE Result	Before SMOTE Churn=0: 4,130 | Churn=1: 1,495	**After SMOTE Churn=0: 4,130 | Churn=1: 4,130 ✓**

6. Machine Learning Models

Three machine learning models were trained using the SMOTE-balanced training data and evaluated on the original (unbalanced) test set to simulate real-world performance. Models were selected to represent a progression from simple to complex algorithms.

1	**Logistic Regression — Baseline** Simple, interpretable linear model. Used as baseline to benchmark more complex models.	**Parameters:** max_iter=1000, random_state=42
2	**Random Forest — Best Model ■** Ensemble of 100 decision trees. Handles non-linear patterns and feature interactions effectively.	**Parameters:** n_estimators=100, max_depth=10, class_weight='balanced'
3	**XGBoost — Advanced** Gradient boosting algorithm. High performance but requires hyperparameter tuning to reach full potential.	**Parameters:** n_estimators=100, learning_rate=0.1, max_depth=5

7. Model Evaluation & Results

Since accuracy alone is misleading with imbalanced data (a model predicting 'No Churn' for everyone would achieve 73% accuracy), five comprehensive metrics were used to evaluate each model. **F1 Score and ROC-AUC were the primary metrics** given the class imbalance.

Metrics Explanation

Metric	What It Measures	Why It Matters for Churn
Accuracy	Overall correct predictions	Not reliable — misleading with imbalanced data
Precision	Of predicted churners, how many actually churned	Reduces false alarms — unnecessary retention offers
Recall	Of actual churners, how many did we catch	Most critical — missing churners means losing customers
F1 Score	Balance between Precision and Recall	Primary metric — best overall performance indicator
ROC-AUC	Model's ability to distinguish churners	Measures ranking quality — higher = better model

Complete Results Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Churners Caught
Logistic Regression	0.7441	0.5132	0.7299	0.6026	0.8209	273 / 374
Random Forest ★	**0.7569**	**0.5301**	**0.7540**	**0.6225**	**0.8338**	**282 / 374**
XGBoost	0.7512	0.5233	0.7219	0.6067	0.8284	270 / 374

★ Random Forest is highlighted as the best performing model across all key metrics

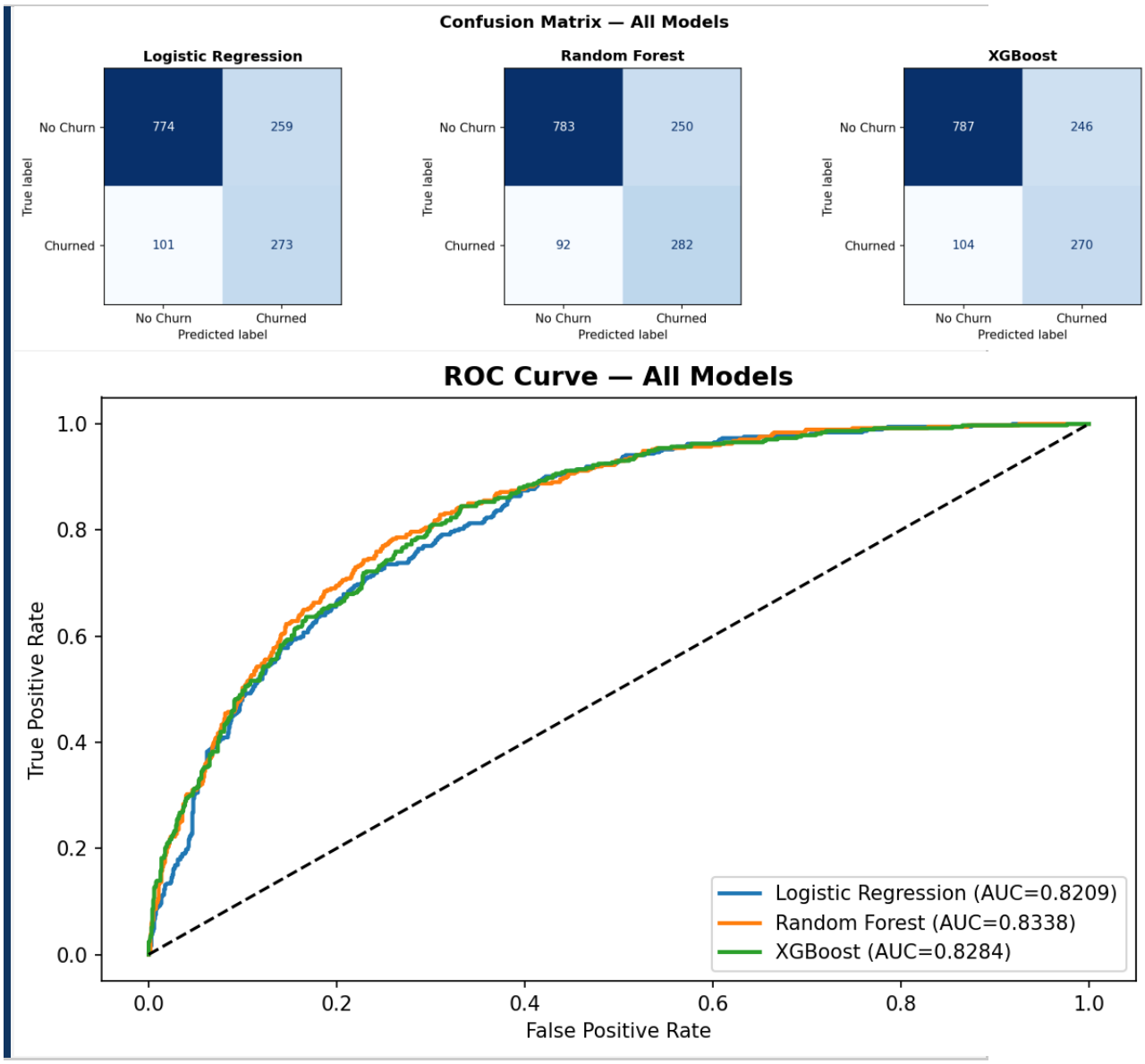


Figure 2: Confusion Matrix and ROC Curve — All Three Models

8. SHAP Explainability Analysis

SHAP (SHapley Additive exPlanations) was applied to the Random Forest model to provide transparent and interpretable explanations for predictions. Unlike traditional feature importance, SHAP explains both **global patterns** (what drives churn across all customers) and **individual predictions** (why a specific customer is predicted to churn).

Top 5 Churn Predictors (SHAP Analysis)

Rank	Feature	SHAP Impact	Direction	Business Meaning
1	tenure	0.085	Negative	Longer tenure = much less likely to churn
2	InternetService_Fiber optic	0.065	Positive	Fiber optic users are high churn risk
3	PaymentMethod_Electronic check	0.055	Positive	E-check users churn more than other methods
4	Contract_Two year	0.050	Negative	Two-year contracts strongly protect against churn
5	TotalCharges	0.040	Negative	Higher total charges = longer customer = lower risk

SHAP Visualizations

Summary Plot (Beeswarm)	Shows global feature importance. Each dot is a customer. Red = high feature value, Blue = low value. Position on X-axis shows impact on churn prediction. Tenure dominates with the widest spread confirming it as the strongest predictor.
Bar Plot	Shows average absolute SHAP values per feature. tenure (0.085) leads, followed by Fiber optic (0.065) and Electronic check (0.055). Confirms contract type and tenure as the two most actionable business levers.
Waterfall Plot	Explains one specific customer's churn prediction. Shows exactly how each feature pushes the prediction above or below the baseline. Red bars push towards churn, blue bars push away from churn.
Dependence Plot	Shows interaction between MonthlyCharges and tenure. High monthly charges combined with low tenure represents the highest churn risk. Long-tenure customers are protected even at high monthly charges.

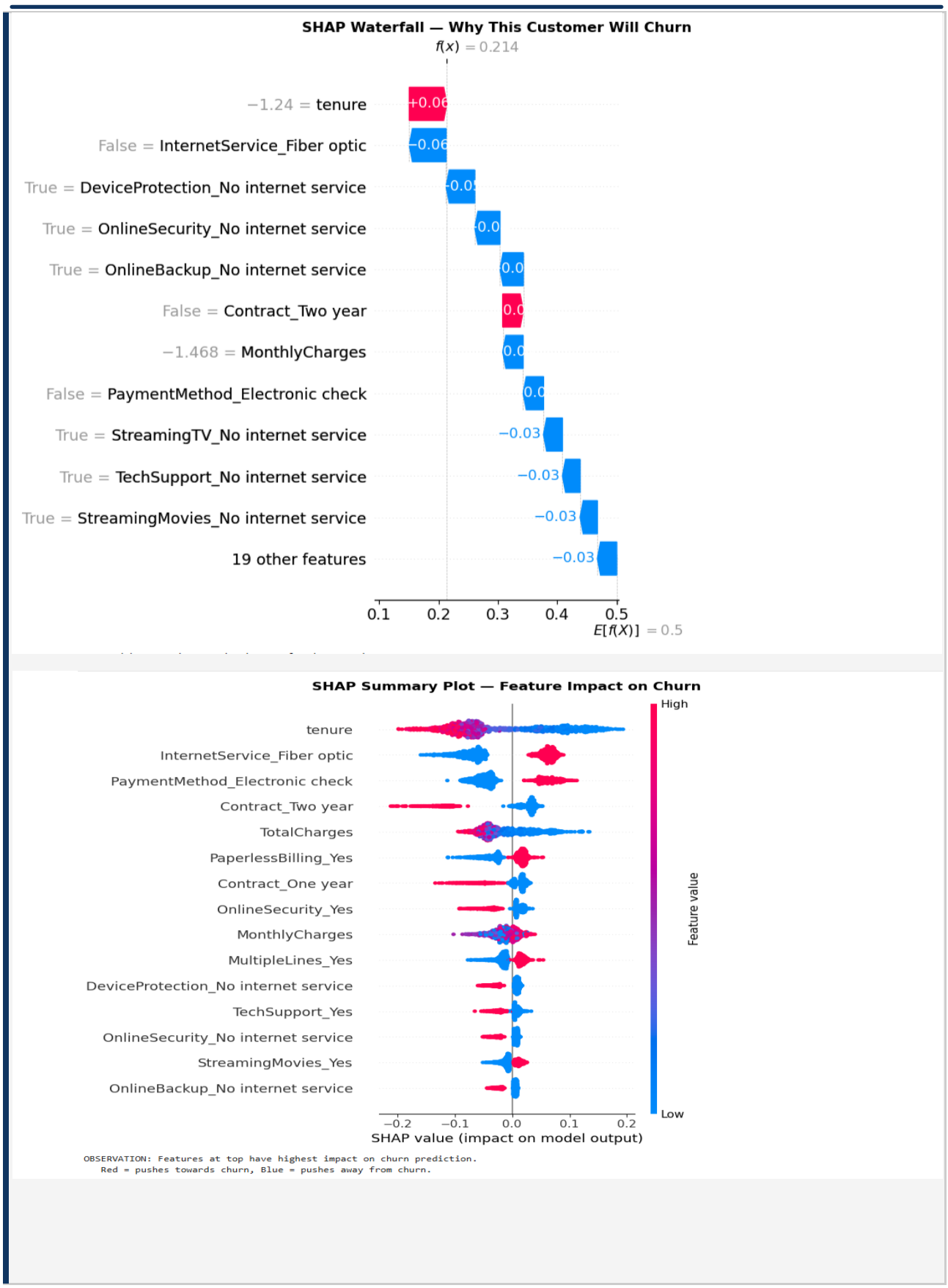
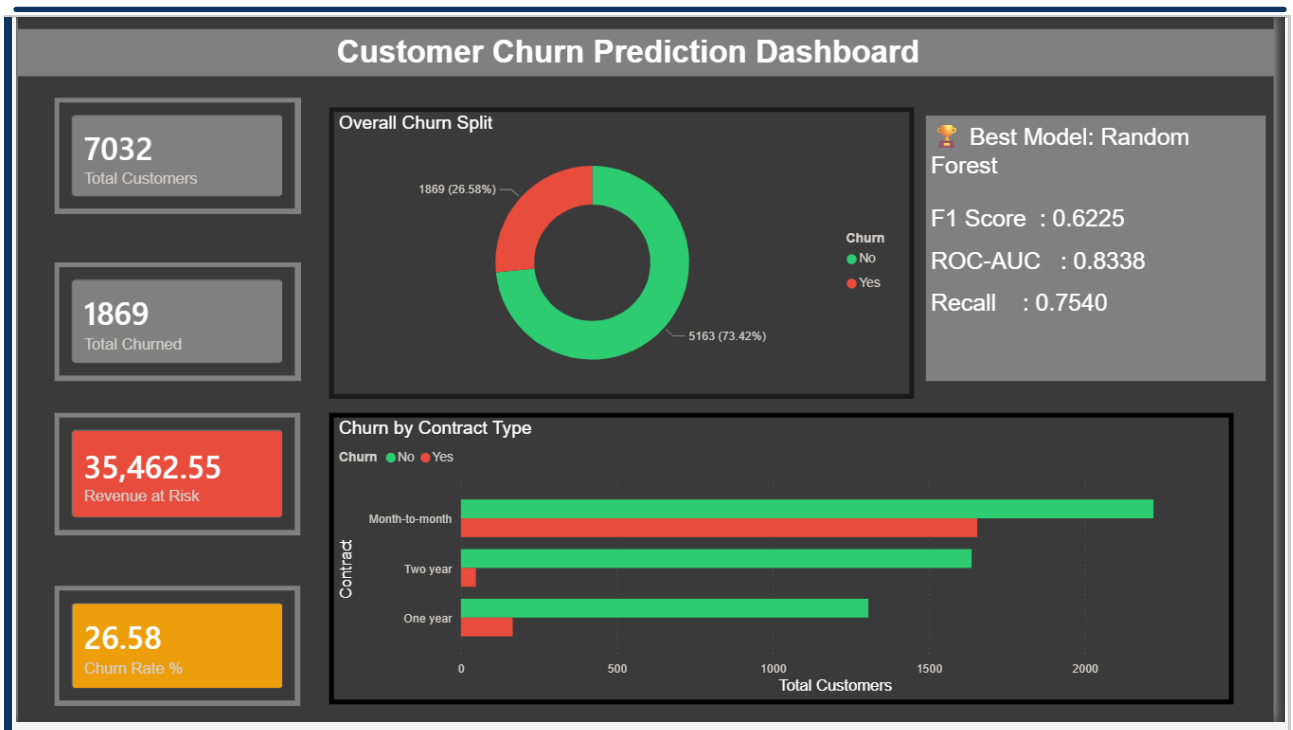


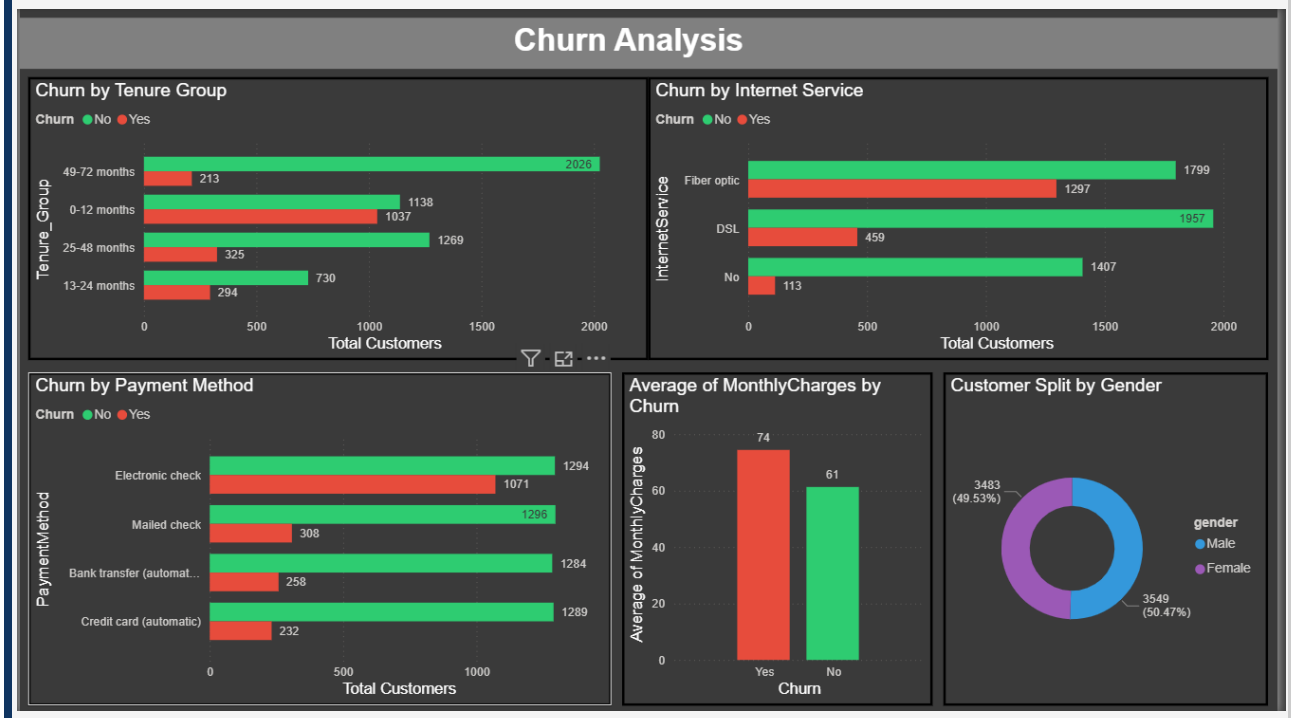
Figure 3: SHAP Analysis — Global Feature Importance and Individual Prediction Explanation

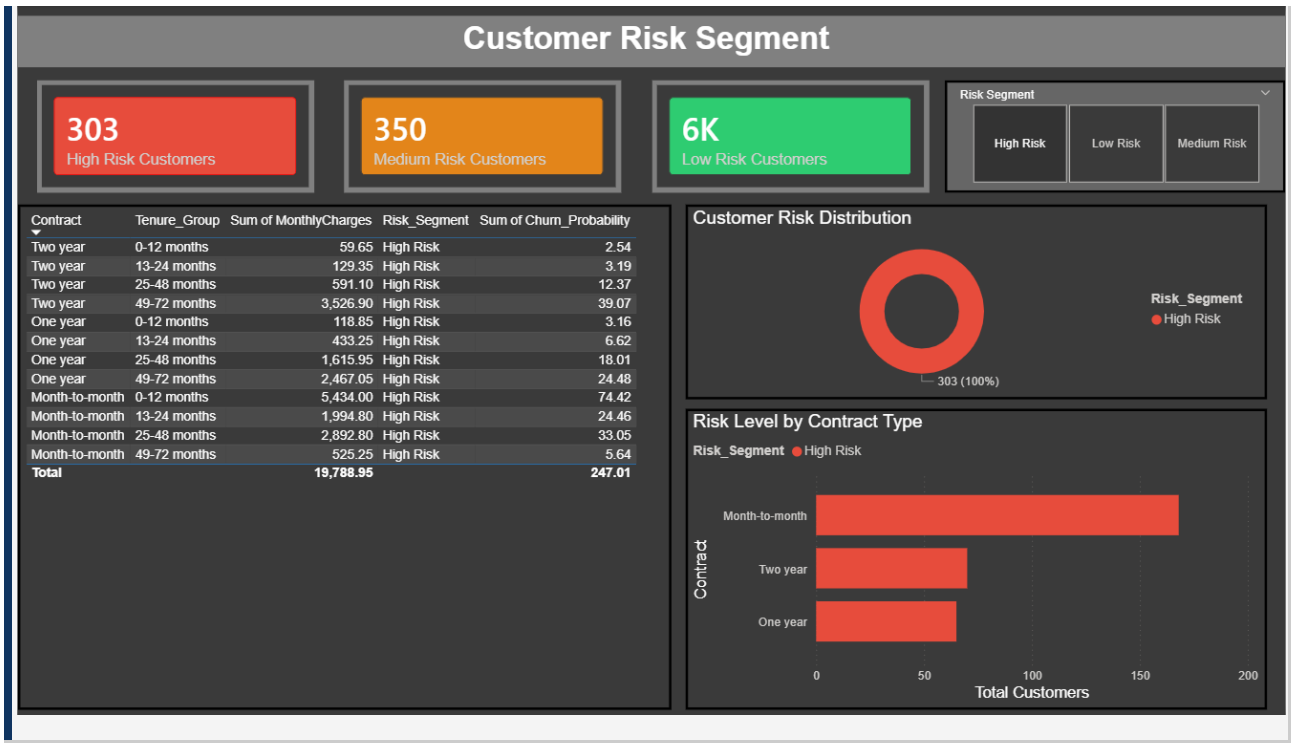
9. Power BI Dashboard

An interactive Power BI dashboard was built to deliver the machine learning insights to business stakeholders in an accessible and visually compelling format. The dashboard comprises three pages, each serving a distinct business purpose.

Page 1	Executive Overview
Visuals: • Total Customers: 7,032 | Churned: 1,869 | Churn Rate: 26.58% | Revenue at Risk: $35,462/month • Overall Churn Donut Chart | Churn by Contract Type Bar Chart • Best Model card — Random Forest (F1: 0.6225, ROC-AUC: 0.8338)	High-level business KPIs for executive decision makers. Shows the scale of the churn problem at a glance.
Page 2	**Churn Analysis**
Visuals: • Churn by Tenure Group | Churn by Internet Service • Churn by Payment Method | Churn by Gender • Interactive filters across all visuals	Deep dive into churn patterns across key customer segments. Helps identify which customer profiles are most at risk.
Page 3	**Customer Risk Segment**
Visuals: • High Risk: 303 customers (Red KPI Card) | Medium Risk: 350 (Orange) | Low Risk: 6,379 (Green) • Risk Distribution Donut Chart | Risk by Contract Stacked Bar • Top 25 At-Risk Customers Table | Risk Segment Slicer	ML-powered risk segmentation. Business teams can filter by risk level and see exactly which customers to target for retention.


Customer Churn Prediction Dashboard
7032
Total Customers
1869
Total Churned
35,462.55
Revenue at Risk
26.58
Churn Rate %
Overall Churn Split
1869 (26.58%)
5163 (73.42%)
Churn
No
Yes
Best Model: Random Forest
F1 Score : 0.6225
ROC-AUC : 0.8338
Recall : 0.7540
Churn by Contract Type
Churn No Yes
Contract
Month-to-month
Two year
One year
0
500
1000
1500
2000
Total Customers


Churn Analysis
Churn by Tenure Group
Churn No Yes
Tenure_Group
49-72 months
2026
213
0-12 months
1138
1037
25-48 months
1269
325
13-24 months
730
294
0
500
1000
1500
2000
Total Customers
Churn by Internet Service
Churn No Yes
InternetService
Fiber optic
1799
1297
DSL
1957
459
No
1407
113
0
500
1000
1500
2000
Total Customers
Churn by Payment Method
Churn No Yes
PaymentMethod
Electronic check
1294
1071
Mailed check
1296
308
Bank transfer (automat...
1284
258
Credit card (automatic)
1289
232
0
500
1000
Total Customers
Average of MonthlyCharges by Churn
Average of MonthlyCharges
80
60
40
20
0
74
61
Yes
No
Churn
Customer Split by Gender
3483
(49.53%)
3549
(50.47%)
gender
Male
Female



Contract	Tenure_Group	Sum of MonthlyCharges	Risk_Segment	Sum of Churn_Probability
Two year	0-12 months	59.65	High Risk	2.54
Two year	13-24 months	129.35	High Risk	3.19
Two year	25-48 months	591.10	High Risk	12.37
Two year	49-72 months	3,526.90	High Risk	39.07
One year	0-12 months	118.85	High Risk	3.16
One year	13-24 months	433.25	High Risk	6.62
One year	25-48 months	1,615.95	High Risk	18.01
One year	49-72 months	2,467.05	High Risk	24.48
Month-to-month	0-12 months	5,434.00	High Risk	74.42
Month-to-month	13-24 months	1,994.80	High Risk	24.46
Month-to-month	25-48 months	2,892.80	High Risk	33.05
Month-to-month	49-72 months	525.25	High Risk	5.64
Total		**19,788.95**		**247.01**

Figure 4: Power BI Dashboard — Executive Overview, Churn Analysis, Customer Risk Segment

Phase 10

Model Saving & Streamlit Deployment

10. Phase 10 — Model Saving & Deployment

Phase 10 completes the project by saving the trained model to disk and deploying it as a fully interactive, corporate-grade web application using Streamlit. This transforms the project from an analysis into a real, usable product — enabling the business team to get real-time churn predictions for any customer without any technical knowledge.

10A. Model Saving — Jupyter Notebook

Two Jupyter cells were added to save all required model files using joblib and test that the saved model loads and predicts correctly.

File Saved	Location	Purpose
churn_model.pkl	C:\ML Project\models\	Trained Random Forest model — main predictor
scaler.pkl	C:\ML Project\models\	StandardScaler — normalizes new input data
feature_cols.json	C:\ML Project\models\	Feature column list — ensures correct input shape

```
Sample Predictions:
  Customer 1:  STAY  (Probability: 2.57%)
  Customer 2:  CHURN  (Probability: 83.02%)
  Customer 3:  STAY  (Probability: 2.74%)
  Customer 4:  STAY  (Probability: 21.38%)
  Customer 5:  STAY  (Probability: 24.13%)
```

Figure 5: Jupyter Notebook — Phase 10 Cell 2: Saved model loaded and tested successfully

Sample Predictions from Saved Model

After loading the saved model, 5 test customers were predicted to validate the save/load pipeline. The results confirmed the model loads and predicts correctly:

Customer	Prediction	Churn Probability	Risk Level
Customer 1	STAY	2.57%	Low Risk ■

Customer 2	**CHURN**	**83.02%**	**High Risk ■■**
Customer 3	STAY	2.74%	Low Risk ■
Customer 4	STAY	21.38%	Low Risk ■
Customer 5	STAY	24.13%	Low Risk ■

Customer 2 correctly identified as HIGH RISK with 83.02% churn probability — model verified working!

10B. ChurnGuard AI — Streamlit Web Application

A corporate-grade Streamlit web application named **ChurnGuard AI — Telecom Customer Intelligence Platform** was built and deployed locally at http://localhost:8501. The app loads the saved Random Forest model and delivers real-time churn predictions through a professional dark-themed interface with 4 fully functional tabs.

Tab	Name	Key Features & Functionality
Tab 1	■ Risk Analysis	Customer name + ID input, prediction banner (CHURN / STAY), churn probability gauge chart, risk factor breakdown bar chart, full customer profile summary, export CSV and text report buttons
Tab 2	■ Analytics Dashboard	Business KPI cards (7,032 customers, 26.58% churn, $35,463 at risk), 4 interactive Plotly charts: Churn by Contract, Churn by Tenure, Model Performance Comparison, Internet Service Donut
Tab 3	■ Batch Prediction	Upload any customer CSV file, predict churn for thousands of customers at once, filter results by risk segment (High/Medium/Low), download full results as CSV
Tab 4	■■ Model Info	Complete project overview, model performance comparison table, top SHAP predictors, full technical stack documentation, project credit

Application Highlights

Dark Corporate Theme Professional navy/red gradient UI — enterprise ready	**Live Risk Gauge Chart** Speedometer showing churn % with color-coded risk zones	**Risk Factor Breakdown** 5 key risk drivers visualized per customer prediction
Batch CSV Processing Predict churn for entire customer base at once	**Export Reports** Download CSV and text reports for each prediction	**Real-time Timestamp** Live date and time header with LIVE badge

Risk Analysis | Analytics Dashboard | Batch Prediction | Model Info

✅ RETENTION LIKELY

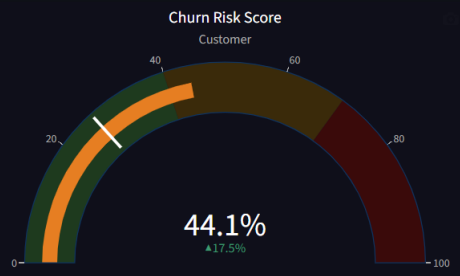
Customer · ID: N/A

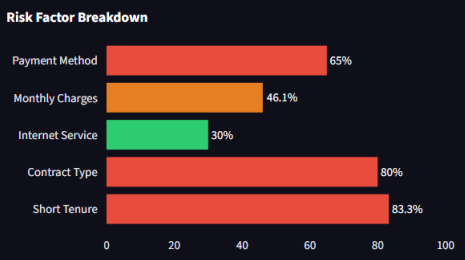
44.1% churn probability

🟠 MEDIUM RISK

Churn Probability	Risk Level	Revenue at Risk	Tenure	Contract
44.1%	MEDIUM RISK	$0	12 months	Month-to-…
↑ 17.5% vs avg			↑ Established	

Deploy ⋮



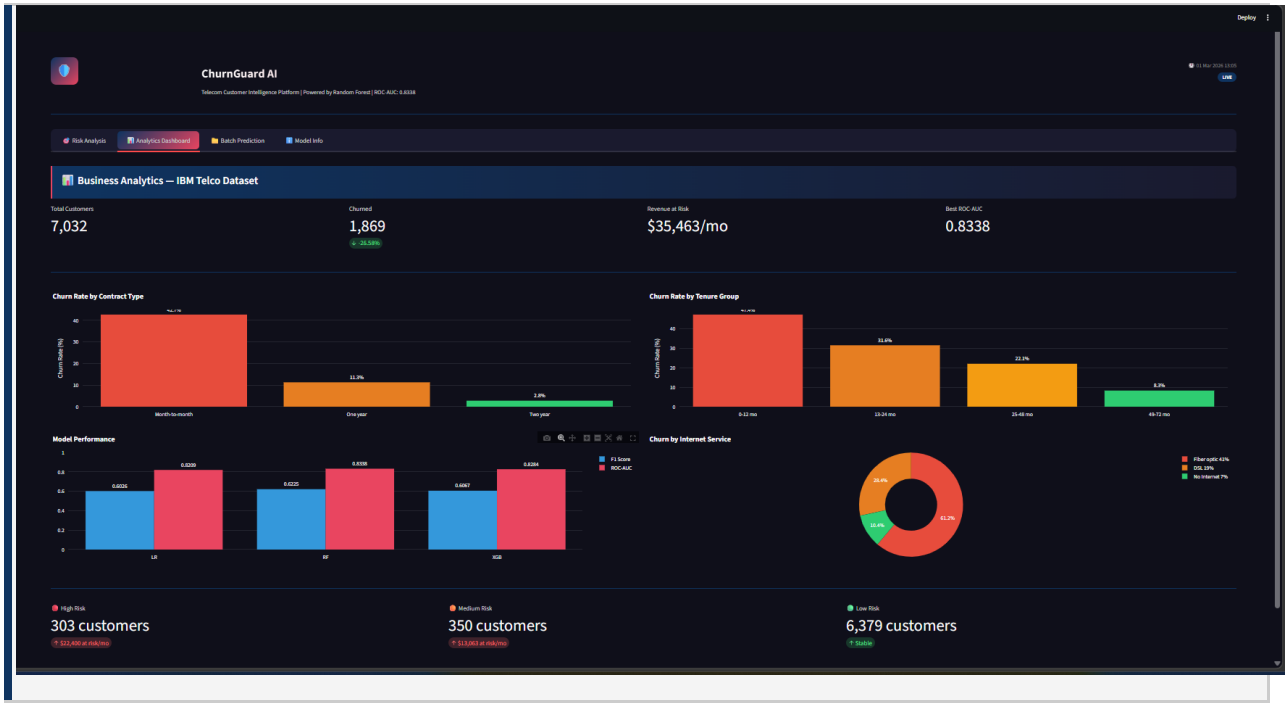


💡 Recommended Action

Proactive monitoring needed. Schedule check-in call and consider loyalty reward.

📋 Complete Customer Profile

👤 **Personal** | 💼 **Account** | 📡 **Services**



How to Run ChurnGuard AI Locally	
Step 1	Open Anaconda Prompt
Step 2	cd "C:\ML Project"
Step 3	streamlit run app.py
Step 4	Browser opens at http://localhost:8501 automatically

■ **Next Step — GitHub + Streamlit Cloud Deployment** The app is ready for public deployment. Files will be pushed to a GitHub repository (churn-prediction-app) and deployed on Streamlit Community Cloud to generate a permanent shareable URL for portfolio, LinkedIn, and resume.

11. Key Insights & Recommendations

Based on the EDA findings, machine learning results, and SHAP explainability analysis, the following high-impact insights and actionable recommendations have been derived for the business to reduce customer churn and protect revenue.

Key Insights

Insight 1	**Contract Type is the #1 Retention Lever**	Month-to-month customers churn at 42% vs 3% for two-year contracts. This 14x difference represents the single most impactful variable in churn prediction.
Insight 2	**New Customers are Highest Risk**	Customers in their first 12 months churn at 47%. The onboarding experience is critical. Losing customers early means the business never recovers the customer acquisition cost.
Insight 3	**Fiber Optic Service Needs Attention**	Fiber optic users churn at 41% vs 19% for DSL users. This suggests pricing, service quality, or competitive pressure is driving dissatisfaction.
Insight 4	**Electronic Check Payment is a Churn Signal**	Customers paying via electronic check churn significantly more than those using automatic payments. This may indicate lower engagement and commitment to the service.
Insight 5	**High Charges Without Commitment = Highest Risk**	SHAP dependence plot confirms that high monthly charges combined with low tenure is the highest-risk customer profile. These customers feel the full cost before experiencing value.
Insight 6	**Revenue at Risk is $35,463 Per Month**	The 303 high-risk customers represent $35,463 in monthly recurring revenue at immediate risk. Even retaining 50% would save over $17,000 per month.

Business Recommendations

Retention Strategy Offer incentives for month-to-month customers to upgrade to one or two-year contracts. Even a 10% conversion rate would significantly reduce churn.	**Onboarding Program** Build a structured 90-day onboarding program for new customers focusing on product education, value demonstration, and early satisfaction checks.
Proactive Outreach Use the Power BI Risk Segment dashboard to identify the top 300 high-risk customers monthly and assign them to a dedicated retention team.	**Payment Method Campaign** Run a campaign encouraging electronic check users to switch to automatic payment methods with a small monthly discount as an incentive.

■ **Fiber Optic Review** Conduct a satisfaction survey specifically for fiber optic users to understand if churn is driven by pricing, performance, or competitive alternatives.	■ **Loyalty Pricing** Introduce loyalty pricing tiers where monthly charges reduce proportionally with tenure to reward long-term customers.

12. Conclusion

This project successfully delivered a complete end-to-end machine learning solution for customer churn prediction. Starting from raw data exploration to a fully deployed interactive web application, every phase was executed systematically following industry best practices.

Phase	Achievement	Status
Phase 1-2	Environment Setup & Data Loading	✓
Phase 3	Exploratory Data Analysis — 6 visualizations, 5 business insights derived	✓
Phase 4	Data Preprocessing — 7-step pipeline including SMOTE balancing	✓
Phase 5	Model Building — Logistic Regression, Random Forest, XGBoost trained	✓
Phase 6	Model Evaluation — 5 metrics, ROC curves, Confusion matrices	✓
Phase 7	SHAP Explainability — 4 SHAP plots, top 5 predictors identified	✓
Phase 8	Power BI Dashboard — 3 interactive business pages built	✓
Phase 10	**Model Saved (joblib) + ChurnGuard AI Streamlit App Deployed**	✓

Business Impact Summary

303 High Risk Customers Identified	282 Churners Caught by Best Model	$35,463 Monthly Revenue at Risk	0.8338 ROC-AUC Best Model

The Random Forest model with SMOTE balancing achieved the best performance with an F1 Score of 0.6225 and ROC-AUC of 0.8338, successfully identifying 282 out of 374 actual churners in the test set. The project has been elevated to a fully deployed product with the ChurnGuard AI Streamlit application, making it immediately usable by business teams for real-time customer risk assessment.

J. CHARAN REDDY

Aspiring Data Scientist

Python | Machine Learning | Power BI | SHAP | Streamlit | Data Analytics